Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Mediterranean Sea and Black Sea Surface Temperature NRT data

SST_MED_SST_L4_NRT_OBSERVATIONS_010_004

SST_BS_SST_L4_NRT_OBSERVATIONS_010_006

SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012

SST_BS_SST_L3S_NRT_OBSERVATIONS_010_013

Issue: 6.0

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Approval date by the CMEMS product quality coordination team: 06/11/2025

CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0 (OSI-TAC)	09/02/2015	All	First version of document	B. Buongiorno Nardelli, A. Pisano, C. Tronconi	
1.1 (OSI-TAC)	05/03/2015	II, III, V	Modified after QUARG review	B. Buongiorno Nardelli, A. Pisano	
1.2 (OSI-TAC)	01/05/2015	All	Change format to fit CMEMS graphical rules. Introduce references to CMEMS.	B. Hackett	L. Crosnier
1.3 (OSI-TAC)	07/03/2017	II	Updated reference sensor list	B. Buongiorno Nardelli	
1.0	25/01/2018	All	Integration of Sentinel-3A and updated reference sensor list	B. Buongiorno Nardelli, A. Pisano, C. Tronconi	
2.0	10/09/2019	All	Integration of Sentinel-3B and updated reference sensor list	B. Buongiorno Nardelli, A. Pisano, C. Tronconi	Mercator Ocean
3.0	16/06/2022		Upstream data change – Integration of VIIRS NOAA20 SST data to be used in the generation of the product in addition to the existing data	Andrea Pisano, Claudia Fanelli, Bruno Buongiorno Nardelli, Cristina Tronconi, Flavio La Padula, Claudia Cesarini.	A. Pisano
4.0	16/06/2023	V.2	Upstream data change – Integration of Metop-C AVHRR SST data. Additionally, a file format change (netcdf version update to 4.1 from 3.6) and the introduction of additional global variables and attributes to be compliant with GHRSST specifications.	Andrea Pisano, Claudia Fanelli, Cristina Tronconi, Flavio La Padula, Claudia Cesarini, Bruno Buongiorno Nardelli.	A. Pisano
5.0	31/05/2024	V.1	Improvement of the effective resolution of the Mediterranean Sea - Ultra High Resolution Sea Surface Temperature Analysis product and enhancement of the accuracy of corresponding SST gradients.	Andrea Pisano, Claudia Fanelli, Daniele Ciani, Cristina Tronconi, Flavio La Padula, Claudia Cesarini, Bruno Buongiorno Nardelli	A. Pisano
6.0	28/05/2025	I.1, II, V, VI, VII, VIII	Upstream data change – Integration of NOAA-21 AVHRR SST data.	Andrea Pisano, Claudia Fanelli, Andrea Massi, Cristina Tronconi, Flavio La Padula, Claudia Cesarini, Bruno Buongiorno Nardelli, Daniele Ciani,	A. Pisano

ACRONYM TABLE

Acronym	Description
AVHRR	Advanced Very High Resolution Radiometer
Argo	International profiling float network (www.argo.net)
BS	Black Sea
CNR-ISMAR	Italian Research National Council – Institute of Marine Sciences
dADR-SR	*dilated Adaptive Deep Residual Network for Super Resolution*
ESA	European Space Agency
GOS	Group of Satellite Oceanography
HR	High spatial Resolution
INS	Copernicus Marine Service In-Situ
MDS	Copernicus Marine Data Store
MED	Mediterranean Sea
MFC	Monitoring and Forecasting Centre
MODIS	Moderate Resolution Imaging Spectroradiometer
MSG	Meteosat Second Generation
MY	Multi-year
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
OI	optimal interpolation
OSI_SAF	Ocean Sea Ice Satellite Application Facilities
PUM	Product User Manual
QC	quality control
REP	Reprocessing
RMSD	root-mean-square difference
SEVIRI	Spinning Enhanced Visible and Infrared Imager
SLSTR	Sea and Land Surface Temperature Radiometer
SST	Sea Surface Temperature
Suomi NPP	Suomi National Polar-orbiting Partnership
TAC	Copernicus Marine Service Thematic Assembly Centre
UHR	Ultra High spatial Resolution
VIIRS	Visible Infrared Imaging Radiometer Suite
XBT	Expendable Bathythermograph

KEY DEFINITIONS

Foundation SST	Foundation Sea Surface Temperature (Foundation SST) refers to the temperature of the uppermost layer of the ocean (typically at a depth of a few centimetres to several meters) that is not affected by diurnal warming or surface cooling due to solar radiation or wind mixing. It represents the baseline or background sea surface temperature, usually estimated during the nighttime or under well-mixed conditions, making it more stable and suitable for climate studies and model assimilation.

RELEVANT DOCUMENT LIST

GHRSST GDS	The Recommended GHRSST Data Specification (GDS) Revision 2.1, June 2022, https://ghrsst.org https://www.ghrsst.org/ghrsst-news/updated-ghrsst-data-specification-2-1-now-available-for-download/
Product User Manual (PUM) for BS NRT SST	Related Product User Manual, CMEMS-SST-PUM-010-006-013
Product User Manual (PUM) for MED NRT SST	Related Product User Manual, CMEMS-SST-PUM-010-004-012

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the four satellite-based Mediterranean Sea and Black Sea (Figure 1) near-real-time (NRT) sea surface temperature (SST) products (Table 1), produced by the Italian Research National Council – Institute of Marine Sciences (CNR-ISMAR) within the Copernicus Marine Service. These products cover the **period from 2008 to present**.

The **Mediterranean Sea (MED) and Black Sea (BS)** are part of the geographical domains of the Copernicus Marine Service. The Mediterranean Sea domain extends across the entire basin up to the Strait of Gibraltar at 5.61°W and includes the Dardanelles Strait, while the Black Sea domain encompasses the entire Black Sea up to the Bosphorus Strait.

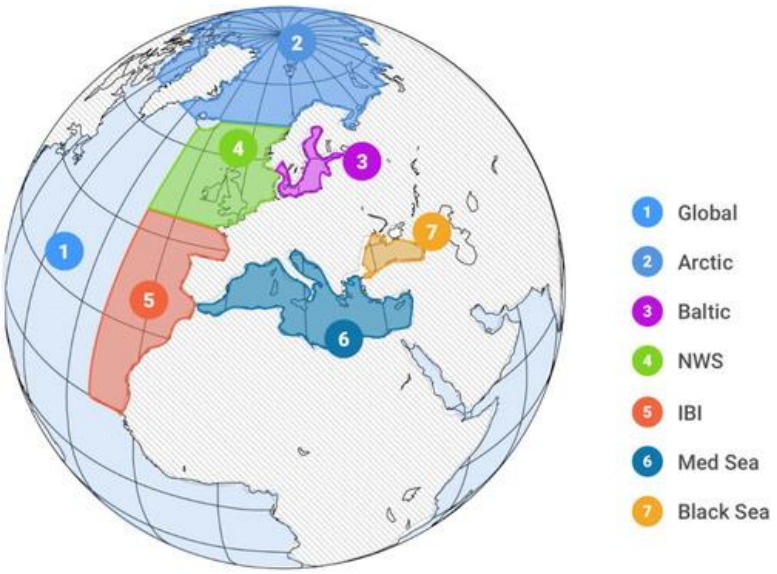

***Figure 1**. Copernicus Marine Service domains. This figure is taken from the Ocean State Report (von Schuckmann et al., 2018). NWS: North West Shelf; IBI: Atlantic-Iberian Biscay Irish.*

Table 1. *The Copernicus Marine Service Mediterranean Sea and Black Sea NRT SST products. Period from 2008 to present.*

PRODUCT REFERENCE	FULL NAME	AREA	TYPE	LEVEL
SST_BS_SST_L3S_NRT_ OBSERVATIONS_010_013	Black Sea - High Resolution and Ultra High Resolution L3S Sea Surface Temperature	BS	NRT	L3S
SST_BS_SST_L4_NRT_ OBSERVATIONS_010_006	Black Sea - High Resolution and Ultra High-Resolution Sea Surface Temperature Analysis	BS	NRT	L4
SST_MED_SST_L3S_NRT_ OBSERVATIONS_010_012	Mediterranean Sea - High Resolution and Ultra High Resolution L3S Sea Surface Temperature	MED	NRT	L3S
SST_MED_SST_L4_NRT_ OBSERVATIONS_010_004	Mediterranean Sea - High Resolution and Ultra High-Resolution Sea Surface Temperature Analysis	MED	NRT	L4

This suite of products provides daily merged multi-sensors (Level-3S, L3S) and optimally interpolated (Level-4, L4) estimates of the foundation SST (i.e., the temperature free of diurnal cycle, Minnett et al., 2019), over the Mediterranean Sea (MED) and Black Sea (BS) domains, at high (HR=1/16°) and ultra-high (UHR=1/100°) spatial resolution. Each product is representative of **night-time SST values** (centred at 00:00 UTC).

L3S data are produced by selecting only the highest quality SST values (to reduce cloud contamination) from lower level (Level-2, L2) upstream data within a short temporal window (at local night-time to avoid diurnal cycle). L2 data that are presently ingested include:

- the Sea and Land Surface Temperature Radiometer (SLSTR) from Sentinel-3A/-3B;
- the Visible Infrared Imaging Radiometer Suite (VIIRS) from NOAA-20, NOAA-21 and SUOMI National Polar-orbiting Partnership (NPP);
- the Advanced Very High Resolution Radiometer (AVHRR) from Metop-B and Metop-C;
- the Moderate Resolution Imaging Spectroradiometer (MODIS) from AQUA and TERRA;
- the Spinning Enhanced Visible and Infrared Imager (SEVIRI).

L3S data are then used as input to the optimal interpolation to derive the L4 data (Buongiorno Nardelli et al., 2013).

Both L3S and L4 data are **produced daily and reprocessed after one month** to guarantee the maximum quality in the archived datasets (using a symmetrical temporal window in the interpolation algorithm).

Additionally, from 2024 onwards, an improved first-guess field has been used in the generation of the MED UHR L4 data, enhancing the product's spatial resolution of SST features and the accuracy of SST gradients via machine learning techniques (Fanelli et al., 2024).

I.2 Summary of the results

The accuracy of both MED and BS NRT L3S and L4 products (Table 1) has been quantified by direct comparison to independent drifter data over a one-year period. The validation period spans from January to December 2018 for the Mediterranean Sea products, and from January to December 2014 for the Black Sea products. The different period is due to the absence of drifters in the Black Sea since 2015.

Drifter data were provided by the Copernicus Marine Service In-Situ (INS) Thematic Assembly Center (TAC). Specifically, mean bias and root-mean-square difference (RMSD) have been computed from the

differences between SST and in-situ data over a matchup dataset (namely, co-located SST and in-situ observations in space and time).

I.3 Estimated Accuracy Numbers

This section includes the estimated accuracy numbers for the products covered in this document.

Specifically, mean bias and root-mean-square difference (RMSD) have been computed from the differences between SST and in-situ data over a matchup dataset (namely, co-located SST and in-situ observations in space and time).

For the Mediterranean Sea products, validation results show a mean bias and RMSD respectively of -0.106 ± 0.006 °C and 0.423 ± 0.006 °C for the HR L4 SST product, and -0.078 ± 0.003 and 0.408 ± 0.003 for the UHR L4 SST product (Table 2). Similar results are found for the MED HR and UHR L3S products. For the Black Sea products (Table 2), the mean bias and RMSD are -0.23 ± 0.08 °C and 0.67 ± 0.09 °C for the HR L4 SST product, and -0.23 ± 0.07 °C and 0.61 ± 0.07 °C for the UHR L4 SST product. Similar values are equally found for the BS HR and UHR L3S data. More details about validation statistics are given in sections III and IV.

Overall, the accuracy of the MED SST data products attains around 0.4 °C, while that of BS data evidences an error around 0.6 °C. However, the statistics for the BS are based on a much lower number of in-situ SST measurements than the Mediterranean Sea for the period considered.

For both MED and BS L4 products, the formal optimal interpolation error (analysis_error variable) appears to slightly overestimate the RMSD of the difference between corresponding interpolated data and in-situ measurements evaluated at fixed bins. However, it also appears to remain almost constant at all interpolation error levels.

Table 2. *Summary of statistics of the Copernicus Marine Service Mediterranean Sea (MED) and Black Sea (BS) SST products at both high (HR) and ultra-high (UHR) spatial resolutions. Mean bias and RMSD are computed from the differences of co-located (matchups') SST and in-situ data from 01-01-2018 to 12-31-2018 and from 01-01-2014 to 12-31-2024 for the Mediterranean Sea and Black Sea SST products, respectively. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron and Tibshirani, 1993). Numbers provided are round to the first significant digit.*

Product/Dataset Name	**CMEMS Area**	**Spatial Resolution**	**Mean bias (°C)**	**RMSD (°C)**
SST_BS_SST_L3S_NRT_OBSERVATIONS_010_013_a (HR)	BS	HR (1/16°)	-0.09 ± 0.13	0.69 ± 0.10
SST_BS_SST_L3S_NRT_OBSERVATIONS_010_013_b (UHR)	BS	UHR (1/100°)	-0.03 ± 0.18	0.61 ± 0.14
SST_BS_SST_L4_NRT_OBSERVATIONS_010_006_a (HR)	BS	HR (1/16°)	-0.23 ± 0.08	0.67 ± 0.09
SST_BS_SST_L4_NRT_OBSERVATIONS_010_006_c_V2 (UHR)	BS	UHR (1/100°)	-0.23 ± 0.07	0.61 ± 0.07
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a (HR)	MED	HR (1/16°)	-0.142 ± 0.007	0.443 ± 0.008
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b (UHR)	MED	UHR (1/100°)	-0.089 ± 0.008	0.420 ± 0.008
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2 (HR)	MED	HR (1/16°)	-0.106 ± 0.006	0.423 ± 0.006
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c (UHR)	MED	UHR (1/100°)	-0.078 ± 0.003	0.408 ± 0.003

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: CNR-ISMAR GOS (National Research Council, Institutes of Marine Research, Group of Satellite Oceanography), Rome (Italy).

Production system names:

- Mediterranean Sea High Resolution and Ultra High Resolution Sea Surface Temperature Analysis (dataset name: SST-MED-SST-L4-NRT-OBSERVATIONS-010-004);
- Black Sea High Resolution and Ultra High Resolution Sea Surface Temperature Analysis (dataset name: SST-BS-SST-L4-NRT-OBSERVATIONS-010-006);
- Mediterranean Sea - High Resolution and Ultra High Resolution L3S Sea Surface Temperature (dataset name: SST-MED-SST-L3S-NRT-OBSERVATIONS-010-012);
- Black Sea - High Resolution and Ultra High Resolution L3S Sea Surface Temperature (dataset name: SST-BS-SST-L3S-NRT-OBSERVATIONS-010-013).

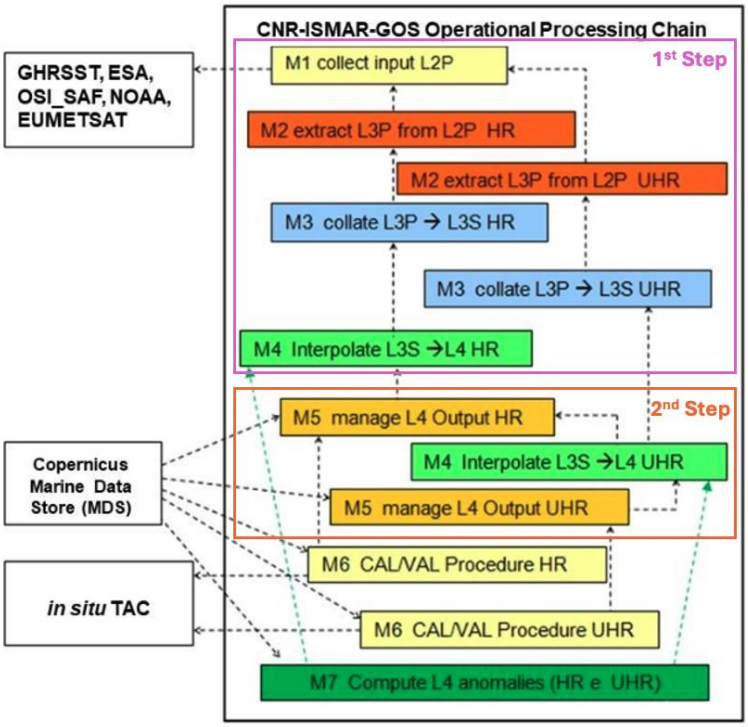

Figure 2. *Schematic diagram of the operational processing chain at CNR-ISMAR-GOS. The whole chain is structured into various modules (referred to as M1, M2, …, M7), which are outlined here as 'logical' steps applicable to both the Mediterranean Sea and Black Sea SST processing. The purple and orange boxes bound together the logical steps associated to the 1st and 2nd algorithm steps outlined in Section II. The arrow's direction follows the UML (Unified Modelling Language) convention (e.g., module M2 needs input from module M1).*

The CNR-ISMAR-GOS SST processing chain is designed as a two-step process (meaning the two-step process that the creation of the L4 UHR is done after the creation of the L4 HR): the high resolution (HR) processing and interpolation is performed first, while the ultra-high resolution (UHR) is run in sequence

(Figure 2). This architecture reflects the specific algorithm used for the UHR interpolation as detailed in Buongiorno Nardelli et al. (2013).

The whole chain is organized in different modules (referred to as M1, M2, …, M7) described here as 'logical' steps, that apply to both the Mediterranean Sea and Black Sea SST processing. These modules are fully described in sections below.

II.1 Data download, data selection and quality control (M1-M2)

The first module (M1) manages the external interface to the input data providers (GHRSST[1], ESA[2], OSI_SAF[3], NOAA[4], EUMETSAT[5]) and the internal input data archive. It includes configurable connection protocols and editable lists of the sensors to be considered. It collects metadata from different international space agencies (i.e., NASA[6], PODAAC[7], EUMETSAT and OSI_SAF) and downloads only the L2P data covering at least a fraction of the interpolation areas. This module is common for both HR and UHR, and it is activated at regular time intervals.

All L2P data are GHRSST-compliant (i.e., follow the GHRSST Data Specifications; GDS[8]) and provided with additional metadata, including quality indicators and cloud flags. The L2P infrared observations currently used to generate the MED SST products are acquired from the following satellite sensors:

- the Sea and Land Surface Temperature Radiometer (SLSTR) on board Sentinel-3A and Sentinel-3B;
- the Visible Infrared Imaging Radiometer Suite (VIIRS) on board NOAA-21, NOAA-20, and Suomi National Polar-orbiting Partnership (Suomi NPP);
- the Advanced Very High Resolution Radiometer (AVHRR) on board Metop-B and Metop-C;
- the Moderate Resolution Imaging Spectroradiometer (MODIS) on board AQUA and TERRA;
- the Spinning Enhanced Visible and InfraRed Imager (SEVIRI) on board the Meteosat Second Generation (MSG) satellite.

Figure 3 illustrates the timeline of sensor integration into the MED SST processing chains from 2018 onwards.

[1] Group for High-Resolution Sea Surface Temperature:. https://www.ghrsst.org/
[2] European Space Agency, https://www.esa.int/
[3] Ocean Sea Ice Satellite Application Facilities, https://osi-saf.eumetsat.int/
[4] National Oceanic and Atmospheric Administration, https://www.weather.gov/
[5] European Organization for the Exploitation of Meteorological Satellites, https://www.eumetsat.int/
[6] National Aeronautics and Space Administration, https://www.nasa.gov/
[7] The Physical Oceanography Distributed Active Archive Center, https://podaac.jpl.nasa.gov/
[8] GHRSST - GHRSST Project Documents. https://www.ghrsst.org/resources/ghrsst-project-documents/

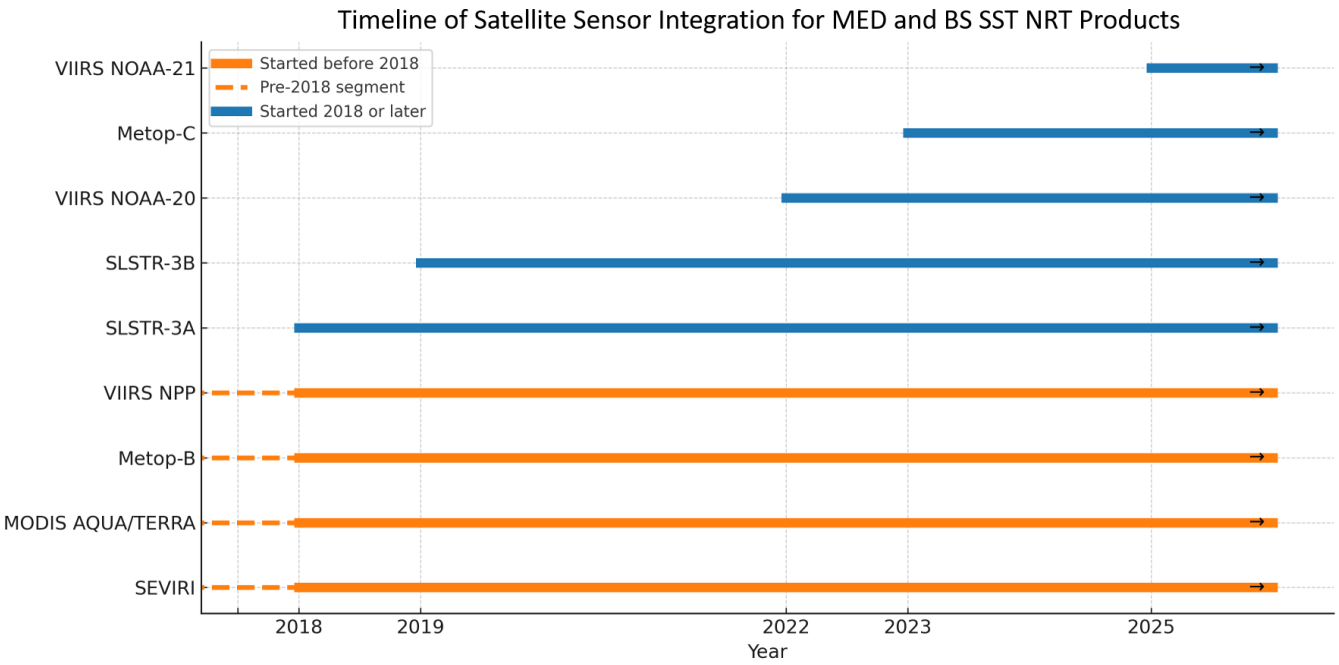

Figure 3. Timeline of sensor integration into the near-real-time SST processing chains for the Mediterranean and Black Seas since 2018.

The second module (M2) is dedicated to data pre-processing. SST observations are initially extracted based on the geographical coverage and the local time of acquisition on a pixel-by-pixel basis. To avoid contamination from diurnal warming effects (Buongiorno Nardelli et al., 2005), the MED SST processing chain selects only those observations collected from 9:00 p.m. on the day before to 6:00 a.m. on the interpolation day (local pixel time).

Subsequently, a quality control step is applied to the L2P data, using the pixel-level quality flags and confidence values. Only observations with the highest quality flag (i.e., flag value equal to 5) are retained. The selected valid data are then remapped onto the final interpolation grids, generating the so-called Level 3 uncollated (referred to as L3P in Figure 2) data. The Mediterranean HR grid was originally designed to match the resolution of the first-level grid used by the Copernicus Marine Service Mediterranean Monitoring and Forecasting Centre (MED MFC), which was 1/16° at the time, to facilitate data assimilation. However, in recent years, the MED MFC grid resolution has been updated to 1/24°.

II.2 Sensor bias adjustment, super-collating and additional cloud screening (M3)

The different L3P images are merged into a single daily super-collated image (L3S). However, the SST estimated from one sensor might significantly differ from that retrieved by another, mainly as a consequence of the differences among the sensors (number of bands, spectral resolution, scanning/viewing geometry, etc.) and/or of the different algorithms applied, which in particular might correct very differently the atmospheric contribution to the measured brightness temperature (see e.g. Minnett et al., 2019). Consequently, a bias adjustment procedure is applied to L3P data before super-collating. Two different algorithms are used in the HR and UHR processing:

- The HR scheme selects the 'best' measure available for each pixel and corrects the biases among the images by adjusting them to a reference SST (which is updated every time that a corrected image is added to the merged map). Here, 'best' is defined through a pre-determined sensor validation statistics, based on which, a simple hierarchy of sensors is identified coherently. SLSTR-3A has been the reference sensor since 2018. During the collating procedure, before adding the new data to the merged map, the large-scale bias between each new image and the pixels that have already been merged is estimated and removed. Additionally, potential cloud contamination is mitigated by flagging pixels that are 1.5 °C colder than the corresponding value in the previous day's L3S product or 5 °C colder than the climatological reference.
- The UHR scheme is based on a different definition of 'best' measure, which keeps into account the continuity of the data present in the single image. The bias is not estimated with respect to the higher accuracy sensor data but between each image and the first guess field which is built directly from the HR L4 SST (see next paragraph. II.3). This bias is estimated and removed locally (50 km). The 'best' data are then selected based on a measure of each image data 'sparseness' (spotty/scattered data being qualified as 'worse').

II.3 Optimal interpolation algorithm (M4)

A classical optimal interpolation (OI) method (Bretherton et al., 1976) has been adapted to the MED and BS SST processing. OI gives an estimate of an anomaly field with respect to a first-guess, assuming the statistical characteristics of the variability are known (background error covariance and observation error covariance). The SST L4 analysis is then obtained as a linear combination of the observations, weighted directly with their correlation to the interpolation point and inversely with their cross-correlation and measurement error. OI also provides an estimate of the interpolation error, basically providing information on the observations impact on each grid point. While several operational systems are based on purely spatial OI algorithms, using daily increments as input data (i.e., using previous day analysis as a first guess and eventually relaxing to a climatology in the absence of new observations) prompts CNR schemes to directly apply a space-time OI.

The MED and BS SST statistical interpolation schemes use different background fields and covariance models depending on the product. The BS HR L4 processing utilizes a daily pentad climatology as first guess, built from 21 years of AVHRR Pathfinder data over the period 1985-2005 (see Marullo et al., 2007). All aspects of this approach (such as the initial estimate calculation, covariance model, data sub-sampling strategy, and inversion technique) are comprehensively detailed in Buongiorno Nardelli et al. (2010).

Similarly, the MED HR L4 processing also relies on a daily pentad climatology for its first guess, derived from the same 21-year AVHRR Pathfinder dataset (1985–2005), as used for the Black Sea. The key elements of the algorithm (including the first-guess computation, covariance model, data sub-sampling strategy, and inversion technique) are detailed in Buongiorno Nardelli et al. (2013). Since April 2014, the algorithm has been upgraded with a new covariance function model, featuring a revised analytical formulation and spatially variable parameters (Buongiorno Nardelli et al.,2015).

The approach used to interpolate the MED and BS SST at UHR is based on a simple concept concerning scales separation/decomposition. In practice, the lower-resolution HR L4 map is first re-binned onto the UHR grid using a simple bilinear interpolation and then used as a first-guess field (Buongiorno Nardelli et al., 2013). Since 2024, for MED UHR, this first-guess has been enhanced by applying a convolutional neural network (CNN) specifically trained to improve both the effective spatial resolution of SST fields and the accuracy of SST gradients. The network used, known as the *dilated Adaptive Deep Residual*

Network for Super Resolution (dADR-SR) was trained to reconstruct fine-scale structures from lower-resolution maps using L3S SST data at 1 km resolution (Fanelli et al., 2024). The super-resolved output of the network is then used as the new first-guess in a second interpolation step, where SST anomalies (now at 1 km resolution) represent only the small-scale signal components.

The background UHR covariance model has been defined as a result of empirical tests. The spatial decorrelation lengths have been varied between 5 and 15 km, but very little differences have been found on the interpolated maps in the whole range considered. The smaller value is currently used in the operational chain. On the opposite, the time scale has been set to 1 day in order to avoid the sudden growth of fake spatial SST gradients in areas differently covered by satellite observations during successive days.

Both HR and UHR schemes however are sub-optimal in the sense that for each interpolation point, the input data are selected only within a limited sub-domain (within a space-time influential radius). In practice, the super-collated L3S data at HR are collected within a temporal window of ten days while at UHR, only the measurements relative to the same day and two previous days are used. The spatial influential radius ranges between 300 and 900 km at HR and is limited to 20 km at UHR.

To avoid data propagation across land, the OI scheme adopts a specific input data selection check. In particular, given the interpolation point and the input point selected within the sub-domain, the latter is considered for the interpolation only if there is no land pixel between the straight line connecting the two points.Before entering the influential data selection within each analysis, super-collated L3 images are checked for residual cloudy pixels at both resolutions. In practice, cloud margins are first eroded, flagging all values within a distance of two pixels to a pixel already flagged as cloudy. A second check is then performed through the comparison to the closer (in time) L4 analysis available, which is used as a reference only if the analysis error is lower than a fixed value.

The interpolation error is obtained from the formal definition of the error variance derived from OI theory (e.g., Bretherton et al., 1976).

II.4 Output data creation, internal validation and anomaly computation (M5-M7)

Module M5 is responsible for producing the L3S and L4 output data in NetCDF format, which is then distributed via the Copernicus Marine Data Store (MDS)[9]. For more information on the data format, please refer to the Product User Manual (CMEMS-SST-PUM-010-004-006 and CMEMS-SST-PUM-010-012-013).

Module M6 provides validation statistics by directly comparing L3S/L4 data with in-situ observations (see the next section for details on validation). Currently, these validation results are not distributed through the MDS.

Finally, module M7 calculates daily SST anomaly fields using a 21-year climatology (1985–2005) based on Pathfinder data (Marullo et al., 2007). These SST anomaly fields are available through the MDS.

[9] https://data.marine.copernicus.eu/products

III VALIDATION FRAMEWORK

This section introduces the validation protocols used to assess the Mediterranean Sea and Black Sea NRT SST products.

The baseline for the validation of SST TAC products is exclusively based on the use of drifting buoy measurements, in accordance with the recommendations of the international SST scientific community, such as those of GHRSST (see also Minnett et al., 2019).. The validation activities carried out at CNR-ISMAR-GOS are based on the compilation of a matchup database between satellite SST products and surface drifting buoys measurements. More specifically, the in-situ SST data used are those made available by the CMEMS In-situ (INS) TAC.

Before computing the statistics of the differences between satellite and in-situ SST estimates, specific quality control (QC) procedures and matchup protocols are applied. In fact, drifting buoy sensors may be affected by significant systematic errors or may experience malfunctions because of a variety of factors (lost drogue, broken sensor, etc.). As a consequence, only the data that passed all QC tests were included in the matchup database. Additional tests include a comparison with the daily climatological SST field that is used to compute the SST L4 anomalies (rejecting all values exceeding a difference of 5°C) and, a comparison with the previous day L4 product (rejecting all values exceeding a difference of 1.5°C which is the same criterion used for satellite L2P data rejection in the CNR-ISMAR-GOS operational NRT L4 processing chain). Further details on our QC protocols and thresholds can be found in Buongiorno Nardelli et al. (2013).

The drifting buoys collect SST data at regular time intervals (based on an internal clock) while position measurements are recorded at longer time intervals, according to the satellite positioning system. However, the matchup has been restricted to night-time data, keeping only one value per buoy. The single SST value per buoy and per day used here corresponds to the measurement collected closest in time and space to the satellite observation..

Formal interpolation errors obtained through the OI have also been compared with matchup SST differences (see section IV).

IV VALIDATION RESULTS

This section shows the validation results for the Mediterranean Sea and Black Sea NRT SST products.

IV.1 Multi-sensor (level 3S) and analysis (level 4) SST products over the Mediterranean Sea (010-012/010-004)

The key statistical parameters (mean bias and RMSD relative to in situ-data) calculated for the Mediterranean L3S and L4 SST products at both high (HR) and ultra-high (UHR) spatial resolutions are shown in Table 3. This assessment uses only drifter data and covers the entire validation period of 2018, spanning from January 1 to December 31.

A mean bias ranging from -0.078 to -0.0142 °C, and a RMSD ranging from 0.408 to 0.443 °C was found. The obtained statistics indicate good agreement between the L3S and L4 SST fields and in situ measurements. The mean bias, ranging from –0.078 to –0.014 °C, shows no significant systematic deviation, while the RMSD values between 0.41 and 0.44 °C confirm that the overall error remains below half a degree, which is consistent with the expected accuracy for high-quality SST products. These results demonstrate the reliability and robustness of the interpolation approach used for generating the SST fields.

***Table 3**. Summary statistics for the Mediterranean SST NRT products. Mean bias and RMSD are computed from the differences between co-located (matchups) SST and in-situ data from 01-01-2018 to 12-31-2018. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Product Name	Spatial Resolution	Mean bias (°C)	RMSD (°C)	Matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a (HR)	HR (1/16°)	-0.142 ± 0.007	0.443 ± 0.008	13151
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b (UHR)	UHR(1/100°)	-0.089 ± 0.008	0.420 ± 0.008	10462
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2 (HR)	HR (1/16°)	-0.106 ± 0.006	0.423 ± 0.006	19004
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2 (UHR)	UHR(1/100°)	-0.078 ± 0.003	0.408 ± 0.003	60014

Figure 4 shows the MED HR and UHR L4 statistics (mean bias and RMSD) as a function of the interpolation error, expressed in °C. Basically, this plot tells us that: (1) when the interpolation error gets close to 0 °C (all observations, no data voids), the corresponding RMSD is about 0.4 °C, which corresponds to the minimum RMSD in the curve; (2) the RMSD remains practically constant (i.e., around 0.45 °C). This result further confirms the robustness of the optimal interpolation algorithm, which can reconstruct SST fields with errors below 0.5 °C even in the presence of significant data gaps.

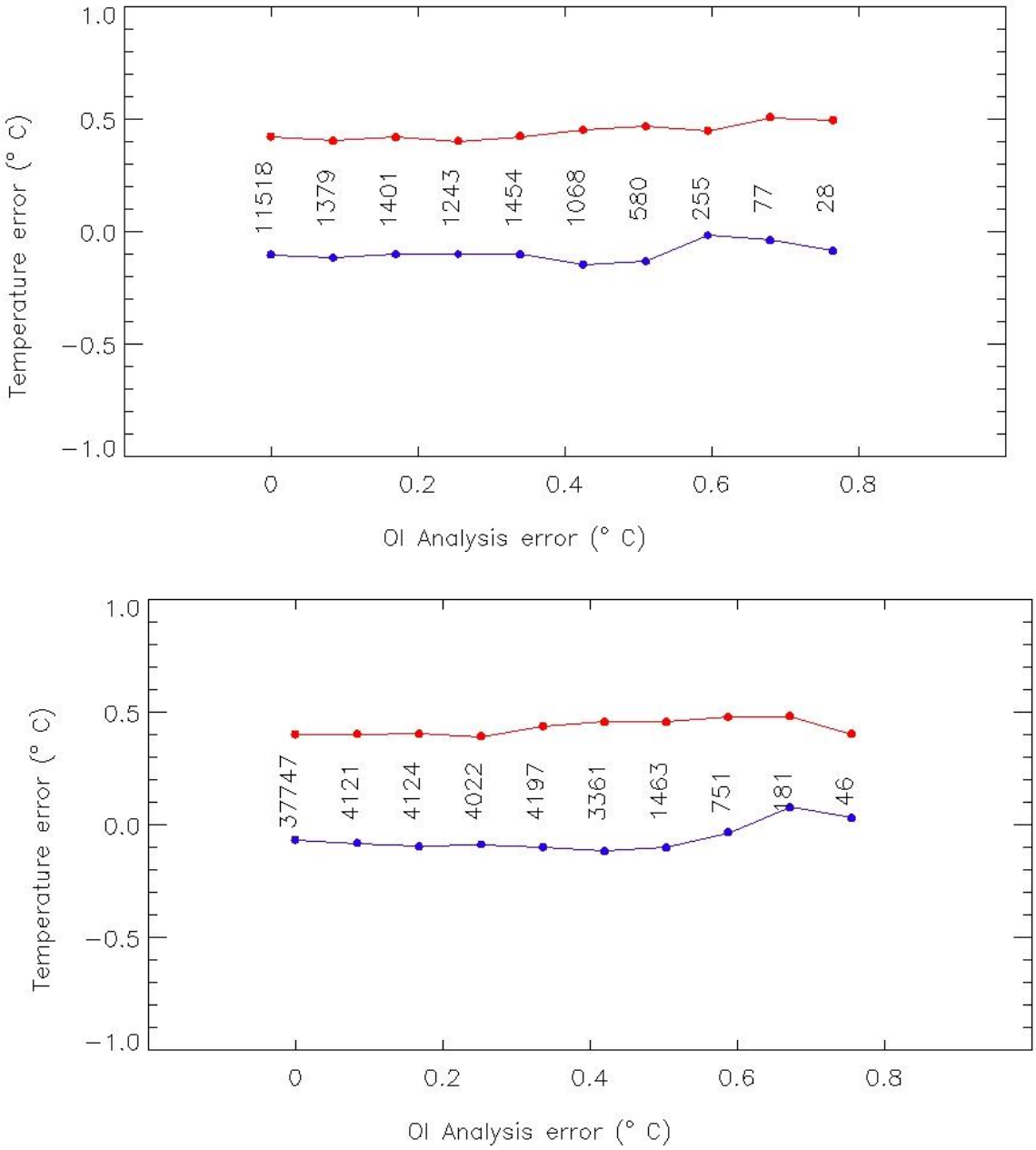

***Figure 4**. Mean bias (blue) and RMSD (red) -y-axis- for the differences between HR (upper panel) and UHR (lower panel) MED Sea L4 and in-situ SST measurements provided by INS TAC as a function of the formal interpolation error -x-axis-. The statistics were computed over the Mediterranean Sea area from 2018-01-01 to 2018-12-31. The number of matchups used to compute the statistics is indicated between the two lines. Units: °C.*

IV.2 Multi-sensor (level 3S) and analysis (level 4) SST products over the Black Sea (010-013/010-006)

Table 4 presents the key statistical parameters—specifically, the mean bias and RMSD relative to in situ data—calculated for the Black Sea L3S and L4 SST products at both high (HR) and ultra-high (UHR) spatial resolutions. This assessment uses only drifter data and covers the entire validation period of 2014, spanning from January 3 (first day of availability for in-situ data) to December 31.

It was found a mean bias ranging from -0.23 to -0.03 °C, and a RMSD ranging from 0.61 to 0.69 °C. These results are slightly higher than those obtained for the Mediterranean Sea, likely due to the limited and sparse availability of in situ data for the Black Sea. Nevertheless, they remain acceptable considering the specific characteristics of the Black Sea, a region highly affected by cloud cover.

***Table 4**. Summary of statistics of the Black Sea SST NRT products. Mean bias and RMSD are computed from the differences of co-located (matchups') SST and in-situ data from 01-03-2014 to 12-31-2014. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993). Numbers provided are round to the first significant digit.*

Product Name	Spatial Resolution	Bias (°C)	RMSD (°C)	Matchups
MSST_BS_SST_L3S_NRT_OBSERVATIONS_010_013_a (HR)	HR (1/16°)	-0.09 ± 0.13	0.69 ± 0.10	107
SST_BS_SST_L3S_NRT_OBSERVATIONS_010_013_b (UHR)	UHR(1/100°)	-0.03 ± 0.18	0.61 ± 0.14	45
SST_BS_SST_L4_NRT_OBSERVATIONS_010_006_a_V2 (HR)	HR (1/16°)	-0.23 ± 0.08	0.67 ± 0.09	276
SST_BS_SST_L4_NRT_OBSERVATIONS_010_006_c_V2 (UHR)	UHR(1/100°)	-0.23 ± 0.07	0.61 ± 0.07	276

Figure 5 illustrates the BS HR and UHR L4 statistical metrics (mean bias and RMSD) relative to the interpolation error, measured in °C. Similar to the Mediterranean analysis, when the interpolation error approaches 0 °C, the RMSD stabilizes around 0.6 °C, representing the minimum on the curve and remains nearly constant at approximately 0.65 °C. However, in contrast to the Mediterranean results, the Black Sea analysis shows a higher bias and RMSD, which - also in this case - can be partly attributed to the fewer available matchup points.

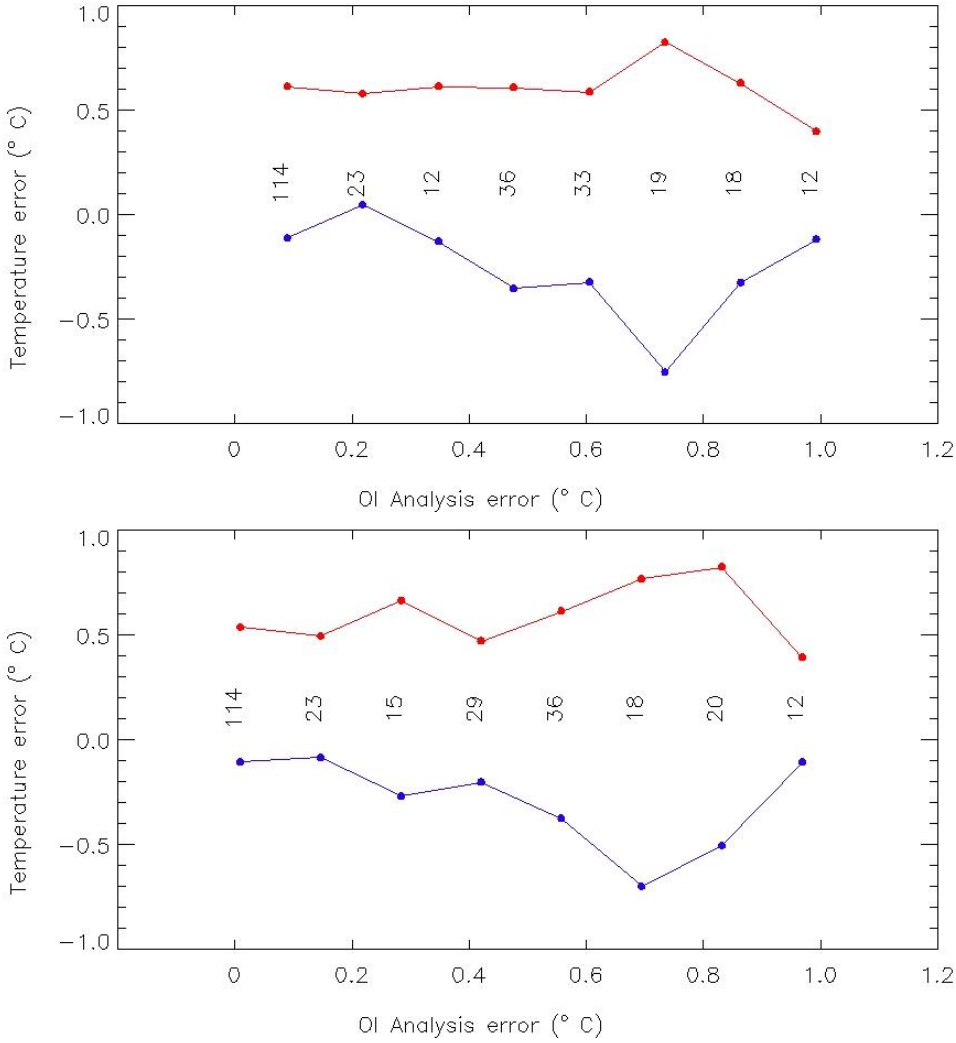

***Figure 5**. Mean bias (blue) and RMSD (red) -y-axis- for the differences between HR (upper panel) and UHR (lower panel) MED Sea L4 and in-situ SST measurements provided by INS TAC as a function of the formal interpolation error -x-axis-. The statistics were computed over the Black Sea area from 01-01-2014 to 12-31-2014. The number of matchups used to compute the statistics is indicated between the two lines. Units: °C.*

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describing and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality (Table 5).

The main changes made to the SST processing chains for the Mediterranean Sea and Black Sea since 2018 are summarized in the table below and are further detailed in the Annexes.

Table 5. *System's noticeable changes.*

Date	Change/Event description	System version	Other
2018/01/28	Integration of Sentinel-3A SLSTR	N.A.	See Section I
2019/09/20	Integration of Sentinel-3B SLSTR	N.A.	See Section I
2022/06/16	Integration of VIIRS NOAA20	N.A.	See Section I
2023/06/16	Integration of Metop-C AVHRR	N.A.	See Section I
2024/05/31	Improvement of the effective resolution of the Mediterranean Sea - Ultra High Resolution Sea Surface Temperature Analysis product and enhancement of the accuracy of corresponding SST gradients through machine learning techniques.	N.A.	See Section I
2025/05/28	Integration of VIIRS NOAA21	N.A.	See Section 0

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version where, for current version, specifically we refer to the integration of AVHRR SSTs from the NOAA-21 satellite.

This section documents the integration of AVHRR SST data from the NOAA-21 satellite (operational since November 2022) into the Mediterranean and Black Sea near-real-time (NRT) processing chains (Section 0). NOAA-21 succeeds NOAA-20, which is approaching the end of its operational life. As this is a like-for-like or comparable replacement, no impact on product quality was expected; nonetheless, a thorough validation was conducted. It is worth noting that NOAA-20 will continue to be used as long as it remains available.

The validation based on one full year (2024) of NOAA-21 AVHRR data revealed no statistically significant differences in mean bias or RMSD when compared to the previous configuration. The accuracy of the updated MED L3S and L4 HR/UHR products was assessed through direct comparison with surface drifting buoys, using the methodology outlined in Section III.

Table 7 presents summary statistics (mean bias and RMSD) for the upgraded products against drifting buoy measurements, while Table reports the same metrics for the previous version without NOAA-21 data. The ingestion of NOAA-21 yielded significant reductions of the mean bias for the MED L4 HR (004_a_V2), the L4 UHR (004_c_V2) and the MED L3S UHR (012_b). It equally yielded a mean bias increase for the MED L3S HR (012_a), although the values are always $O(10^{-2}$ °C). Conversely, the RMSDs did not yield statistically significant changes, except for the MED L4 UHR (004_c_V2), where a small, yet significant reduction in the mean bias is observed. As a result, NOAA-21 has been successfully integrated into both the MED and BS operational systems. However, the impact in the Black Sea could not be validated due to the absence of drifting buoys in the region since 2015.

***Table 6**. Validation results for the MED L3S/L4 HR/UHR SST products upgraded with NOAA-21. Validation period from January to December 2024. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Product/Dataset Name (Upgraded with VIIRS-N21)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.023 ± 0.002	0.397 ± 0.002	201051
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	0.014 ± 0.002	0.384 ± 0.002	173103
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	-0.003 ± 0.002	0.389 ± 0.002	243085
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.004 ± 0.002	0.394 ± 0.002	242650

***Table 7**. Validation results for the MED L3S/L4 HR/UHR SST products without NOAA-21 AVHRR. Validation period from January to December 2024. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Product/Dataset Name (Previous Version)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.012 ± 0.002	0.398 ± 0.002	193367
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	0.022 ± 0.002	0.387 ± 0.002	173412
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	0.006 ± 0.002	0.397 ± 0.002	243186
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.011 ± 0.002	0.401 ± 0.002	242993

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VIII ANNEXES

This section illustrates the evolutions listed in Section V

VIII.1 Feature resolution improvement in the MED UHR NRT product

In 2024, the MED UHR SST NRT processing chain underwent an evolution aimed at enhancing the feature resolution of the generated product. Previously, the optimal interpolation algorithm used to fill data gaps relied on an upscaled version of the L4 HR field as the initial guess. This first-guess field remained unchanged in the absence of L3 UHR data (Buongiorno Nardelli et al., 2013), inherently limiting the effective resolution of the resulting L4 UHR product to below its nominal 1/100° resolution.

To address this, a convolutional neural network called the Dilated Adaptive Deep Residual Network for Super-Resolution (Buongiorno Nardelli et al., 2022) was trained to restore the small-scale features smoothed out in the initial guess before it is used as a background field in the UHR OI scheme (Fanelli et al., 2024). The network was trained to enhance the effective resolution of the initial guess fields using a ground-truth, super-collated L3S UHR SST dataset, specifically constructed from Sentinel-3A and Sentinel-3B SLSTR acquisitions.

The validation of the final MED L4 UHR product was conducted over a full year (from 01 January 2021 to 31 December 2021) by direct comparison with surface drifting buoys. Table A 1 summarizes the statistics of the differences between the upgraded L4 product and drifting buoy measurements in terms of mean bias and root-mean-square difference (RMSD). For comparison, the same statistics for the previous version (see Section V for previous versions) of the product are also presented in Table A 1. In this case, the upsized L4 HR was used as the background initial guess field for the OI UHR scheme. Overall, the upgraded product shows a slight improvement in mean bias, while the RMSD remains unchanged compared to the previous version.

Table A 1. Validation results for the MED L4 UHR SST product upgraded with ML technique and the previous version. Validation has been performed by direct comparison with surface drifting buoys covering the period from January to December 2021. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).

Upgrade	Product/Dataset Name	Mean Bias (°C)	RMSD (°C)	#matchups
Non-upgraded	SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.052 ± 0.003	0.299 ± 0.003	32722
Upgraded with ML techniques	SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.050 ± 0.003	0.299 ± 0.003	32674

VIII.2 Integration of Metop-C AVHRR SST

This section describes the integration of AVHRR SST data from the Metop-C satellite (which became operational in November 2023) into the Mediterranean and Black Sea near-real-time (NRT) processing chains (Section 0). Incorporating newly available sensors is essential to ensure that the MED and BS NRT production remains up-to-date while preserving high quality and accuracy.

Prior to operational integration, a comprehensive validation was performed using one full year (from 01 January to 31 December 2022) of AVHRR Metop-C data (hereafter referred to as Metop-C). The results showed no statistically significant differences in mean bias or RMSD compared to the previous configuration.

In particular, the accuracy of MED L3S/L4 HR/UHR data upgraded with Metop-C was assessed through direct comparison with surface drifting buoys, following the same strategy described in section III. Table A 2 shows summary statistics in terms of mean bias and RMSD of the difference between the upgraded L3S/L4 products, and drifting buoy measurements. As comparison with the previous version, Table A 3 shows the same summary statistics but obtained using the processing chains without Metop-C.

With respect to the previous version, the integration of Metop-C increases the number of input observations as shown by the number of matchups, but does not change either the mean bias or the RMSD. Thus, Metop-C has been integrated in the MED processing chains and in the BS operational systems as well. However, it was not possible to validate the impact of Metop-C against drifting buoys as drifters have been missing in the Black Sea since 2015.

Table A 2. Validation results for the MED L3S/L4 HR/UHR SST products upgraded with AVHRR Metop-C. Validation period from January to December 2022. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).

Product/Dataset Name (Upgraded with VIIRS-N20)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.06 ± 0.01	0.26 ± 0.01	2700
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	-0.02 ± 0.01	0.24 ± 0.01	2127
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	0.02 ± 0.01	0.25 ± 0.01	4732
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.06 ± 0.01	0.29 ± 0.01	4868

Table A 3. Validation results for the MED L3S/L4 HR/UHR SST products without AVHRR Metop-C. Validation period from January to December 2022. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).

Product/Dataset Name (Previous Version)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.05 ± 0.01	0.25 ± 0.01	2594
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	-0.02 ± 0.01	0.22 ± 0.01	2056
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	0.02 ± 0.01	0.24 ± 0.01	4732
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.06 ± 0.01	0.28 ± 0.01	4861

VIII.3 Integration of NOAA-20 AVHRR SST

This section describes the integration of AVHRR SST data from the NOAA-20 satellite (which became operational in November 2022) into the Mediterranean and Black Sea near-real-time (NRT) processing chains (Section 0). The integration of NOAA-20 followed a careful assessment through dedicated processing tests and inter-comparison exercises, as performed with the other satellite sensor integration described in previous sections. The validation was carried out over 1 year (From 01 January to 31 December 2021) of VIIRS-N20 data and evidenced with respect to the previous version, a reduction in terms of bias and no statistically significant differences in terms of RMSD.

In particular, the accuracy of MED L3S/L4 HR/UHR data upgraded with VIIRS-N20 was assessed through direct comparison with surface drifting buoys, following the same strategy described in section III. Table A 4 shows summary statistics in terms of mean bias and RMSD of the difference between the upgraded L3S/L4 products, and drifting buoy measurements. As comparison with the previous version (see Section V for previous versions), Table A 5 shows the same summary statistics but obtained using the processing chains without VIIRS-N20. With respect the previous version, the integration of VIIRS-N20 decreases the bias which is now practically null, while it does not change the RMSD. Thus, VIIRS-N20 has been integrated in the MED processing chains and in the BS operational systems as well. However, it was not possible to validate the impact of VIIRS-N20 against drifting buoys as drifters have been missing in the Black Sea since 2015.

Table A 4. Validation results for the MED L3S/L4 HR/UHR SST products upgraded with VIIRS-N20. Validation period from January to December 2021.

Product/Dataset Name (Upgraded with VIIRS-N20)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.03 ± 0.01	0.31 ± 0.01	10455
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	0.02 ± 0.01	0.30 ± 0.01	8749
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	-0.01 ± 0.01	0.32 ± 0.01	13931
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.02 ± 0.01	0.30 ± 0.01	13896

Table A 5. Validation results for the MED L3S/L4 HR/UHR SST products without VIIRS-N20. Validation period from January to December 2021.

Product/Dataset Name (Previous Version)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	0.02 ± 0.01	0.31 ± 0.01	8988
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	0.07 ± 0.01	0.30 ± 0.01	6162
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	0.06 ± 0.01	0.33 ± 0.01	13919
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.07 ± 0.01	0.32 ± 0.01	13886

VIII.4 Integration of SLSTR-3B

SLSTR SST from Sentinel-3B have been integrated in the Mediterranean and Black Sea operational chains since the end of 2019. A careful validation was carried out as follows : SLSTR-3B L2P SST data (v.1.18, available since the 12th of March 2019) have been used to build daily (nighttime) merged multi-sensor (L3S) and gap-free (L4) SST maps for the Mediterranean Sea by using the MED SST operational system and the results have been validated by using drifter observations.

The validation period covered four months (from March 2019, beginning of SLSTR-3B availability, to June 2019) and was performed over the Mediterranean Sea only, since no drifters were available in the Black Sea.

In order to evaluate the quality of SLSTR-3B data, collated single-sensor files (L3C) from SLSTR-B data were built over the Mediterranean Sea, from March to June 2019. As for SLSTR-3A, which is currently the reference sensor in the processing chains, SLSTR-3B data were split into the Dual and Nadir components,and treaded separately. Then, a matchup dataset of co-located S3B L3C and drifter measurements was built, from which the mean bias (L3C minus drifter measurement) and root-mean-square deviation (RMSD) have been computed. The quality of S3B data has been quantified in a bias of 0.11 ± 0.02 °C and a RMSD of 0.44 ± 0.03 °C for the Dual component, and a bias of 0.06 ± 0.02 and RMSD of 0.46 ± 0.02 for the Nadir component. In addition, as inter-comparison exercise, the same analysis has been performed with L3C files from SLSTR-3A, VIIRS, METOP, MODIS AQUA and MODIS TERRA (i.e. all the sensors ingested in the operational chains). Results are shown in Table A 6, indicating that the quality of SLSTR-3B data is quite satisfactory (being less than 0.5 °C).

Table A 6. Summary of collated single-sensor (L3C) product accuracy over the Mediterranean Sea, from March to June 2019

MED HR L3S	**Mean Bias (°C)**	**RMSD (°C)**	**#matchups**
METOP_B	0.05 ± 0.02	0.41 ± 0.02	1666
MODIS_A	-0.16 ± 0.01	0.46 ± 0.02	3620
MODIS_T	0.08 ± 0.01	0.42 ± 0.02	3268
SLSTR-3A Dual	0.11 ± 0.02	0.39 ± 0.02	1546
SLSTR-3A Nadir	0.01 ± 0.02	0.41 ± 0.02	2695
SLSTR-3B Dual	0.11 ± 0.02	0.44 ± 0.03	1162
SLSTR-3B Nadir	0.06 ± 0.02	0.46 ± 0.02	1975
VIIRS_NPP	-0.11 ± 0.01	0.41 ± 0.01	4236

Then, SLSTR-3B L3C data have been used to build merged multi-sensor (L3S) and optimally interpolated (L4) data over the Mediterranean Sea. The accuracy of L3S and L4 MED data has been assessed through a matchup dataset, built following the same strategy described above. Table A 7 shows summary statistics in terms of mean bias and RMSD of the difference between the upgraded L3S/L4 products (i.e. SLSTR-3B integrated as last sensor) and drifting buoy measurements.

In comparison with the previous version, Table A 8 shows the same summary statistics but obtained using the SST MED processing chains without SLSTR-3B. The integration of SLSTR-3B does not affect the accuracy of the L3S/L4 HR/UHR MED products, which remains unchanged with respect to the previous version. Thus, SLSTR-3B has been integrated in the MED processing chains and in the BS operational systems as well. However, it was not possible to validate the impact of SLSTR-3B against drifting buoys as drifters have been missing in the Black Sea since 2015.

Table A 7. Validation results for the MED L3S and L4 SST upgraded products (SLSTR-3B integrated as last sensor). Validation period from March to June 2019.

Product/Dataset Name (Upgraded with SLSTR-3B)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.027 ± 0.01	0.42 ± 0.01	7032
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	0.02 ± 0.01	0.42 ± 0.01	5953
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	-0.026 ± 0.008	0.400 ± 0.008	9508
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	0.006 ± 0.004	0.387 ± 0.005	31193

Table A 8. Validation results for the MED L3S and L4 SST products without SLSTR-3B. Validation period from March to June 2019.

Product/Dataset Name (Previous Version)	Mean Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.03 ± 0.01	0.41 ± 0.01	6964
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	0.02 ± 0.01	0.41 ± 0.01	5797
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	-0.034 ± 0.008	0.398 ± 0.008	9510
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	-0.002 ± 0.004	0.385 ± 0.004	31247

VIII.5 Integration of SLSTR-3A

SLSTR SST from Sentinel-3A have been integrated in the Mediterranean Sea and Black Sea operational chains in 2018. A careful validation was carried out for the following assessment: Sentinel-3A SLSTR L2P SST data (v.2.18, available since the 5th of July 2017) have been used to build daily (nighttime) merged multi-sensor (L3S) and gap-free (L4) SST maps for the Mediterranean Sea by using the MED SST operational system and, the results have been validated by using drifter observations. The dual and nadir views have been treated separately, primarily because the dual view provides greater accuracy than the nadir view.

The validation period covered six months (from July 2017, beginning of SLSTR availability, to December 2017). SLSTR has been tested in different hierarchy configurations to build the L3S/L4 files (i.e. as reference (first) sensor and last sensor). The matchup dataset was built following the same strategy described in previous sections.

The results indicate that the highest accuracy of the output products is achieved when using the dual view of SLSTR-3A as the reference sensor, while the nadir view is reserved for bias adjustment during the merging procedure (see section II). Table A 9 shows summary statistics in terms of bias and standard deviation of the difference between the upgraded L3S/L4 products (i.e., SLSTR integrated as reference sensor) and drifting buoy measurements. In comparison with the previous version, Table A 10 shows the same summary statistics but obtained using the MED SST processing chain without SLSTR.

Table A 9. Validation results for the MED L3S and L4 SST upgraded products (SLSTR integrated as reference sensor). Validation period from July to December 2017.

Product/Dataset Name (Upgraded with SLSTR)	Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.28 ± 0.01	0.44 ± 0.01	5920
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	-0.24 ± 0.01	0.44 ± 0.01	4947
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	-0.24 ± 0.01	0.43 ± 0.01	7484
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	-0.196 ± 0.005	0.420 ± 0.006	23472

Table A 10. Validation results for the MED L3S and L4 SST products without SLSTR. Validation period from July to December 2017.

Product/Dataset Name (Previous Version)	Bias (°C)	RMSD (°C)	#matchups
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_a	-0.27 ± 0.01	0.44 ± 0.01	5855
SST_MED_SST_L3S_NRT_OBSERVATIONS_010_012_b	-0.22 ± 0.01	0.45 ± 0.01	4733
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_a_V2	-0.21 ± 0.01	0.45 ± 0.01	7511
SST_MED_SST_L4_NRT_OBSERVATIONS_010_004_c_V2	-0.180 ± 0.006	0.432 ± 0.005	23525

SLSTR has been integrated as reference sensor in the BS operational system as well. However, it was not possible to validate the impact of SLSTR against drifting buoys as drifters have been missing in the Black Sea since 2015.